

Practical: Transformer-Based Approaches for Extremes

Edinburgh Summer School on Generative AI for Extremes

THE UNIVERSITY of EDINBURGH

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Dates	10th September 2026
Github Repository	https://github.com/gamex-network/gamex_summer_school_cascades
Software	Python 3.11+; PyTorch 2.10+; JupyterLab; NumPy; SciPy; Matplotlib; pandas
Runtime	2 hours

Materials	01_cascading_extremes_autoregressive.ipynb 02_gamex_continuous_transformer.ipynb 03_gamex_hawkes_cascades.ipynb 04_gamex_cascade_estimation.ipynb
Before the session	Please bring a laptop with Python and the required packages installed; run JupyterLab once to check the environment.
Output	A decoder-only autoregressive Transformer with extremal tokens, alongside a continuous Transformer for cascades of extreme events based on marked Hawkes processes.

Session Aims

By the end of this tutorial, participants should be able to:

- **Sequences of Extremes I**

1. Tokenize the extremal indicator of the continuous multivariate stochastic process into categorical token process containing dynamic cascading extreme event.
2. Construct a decoder-only autoregressive transformer that takes sequence of extremal tokens as input and outputs conditional probability of the next token to appear.

- **Sequences of Extremes II**

1. Adapt the decoder-only Transformer from categorical tokens to continuous data.
2. Explain how a marked Hawkes process makes every event an extreme event, and estimate its parameters and its limit set by EM and maximum likelihood when the genealogy is hidden.
3. Construct and assess a Transformer for cascades whose heads keep the Hawkes closed forms, compare its genealogy inference with the classical estimates, and use it to predict and generate cascades.

Recommended Reading

- The session on *Neural Generative Models in Practice* by Johnny Lee and the school itself [3]
- Hawkes models and their applications by P. J. Laub, Y. Lee, P. K. Pollett and T. Taimre [4]
- Statistical inference for multivariate extremes via a geometric approach by J. L. Wadsworth and R. Campbell [8]
- An Introduction to Statistical Modeling of Extreme Values by S. Coles [2].

Software Setup

A local JupyterLab installation is the recommended setup. The notebooks select CUDA when it is available and otherwise use CPU; a CPU is sufficient, as the largest model has about 120,000 parameters.

```
python -m venv gamex
source gamex/bin/activate          # Windows PowerShell: .\gamex\Scripts\Activate.ps1
python -m pip install --upgrade pip
python -m pip install torch numpy scipy pandas matplotlib jupyterlab
```

Download the code from the repository or, if you have Git installed (optional), clone the repository,

```
git clone https://github.com/clementeferrer/GAMEX_summer_school_cascades.git
cd GAMEX_summer_school_cascades
```

Required Packages

All packages are installed by `requirements.txt`. Every function and model used in the class is written out in the notebook that uses it.

Language	Packages	Use
Python	numpy, scipy, matplotlib	Hawkes simulation, EM and gauge likelihoods, quantile-regression linear programme, GARCH filter, and plotting.
Python	pandas, torch	Event tables; the cascade transformer and its neural gauge.
Notebook	jupyterlab	Running and editing the notebooks.
Optional	yfinance	Only to refresh the cached price file.

Data & Files

The stock market data reads `data/indices.csv`, which ships with the repository, so there is no external download.

Sequences of Extremes I	01_cascading_extremes_autoregressive.ipynb - a 5th order vector autoregressive process is generated.
Sequences of Extremes II	02_continuous_transformer.ipynb - a Gaussian head and sampling, on S&P 500 daily returns. 03_hawkes_cascades_of_extremes.ipynb - the process whose every event is extreme, its simulation, and the classical estimates. 04_transformer_for_hawkes_cascades.ipynb - a Transformer that predicts the next extreme event; genealogy, prediction, generation.
Data	data/indices.csv - daily closes of the three indices.
Workshop runtime	Approximately 2 hours; the extension exercises are optional if time is short.

Key Ideas

Let $\{\mathbf{Y}\}_{t=1}^N$ with $\mathbf{Y} = (Y_{t,1}, \dots, Y_{t,d})$ be a d -dimensional multivariate stochastic process and define the *extremal vocabulary* as $\mathcal{V} = \{0, 1, \dots, d\}$. Then, a sequence of *extremal tokens*, $\mathcal{A} = (A_1, \dots, A_N)$ with $A_t \in \mathcal{V}$ is defined through the probability-scale scoring rule,

$$A_t = \begin{cases} 0, & \mathcal{E}_t = \emptyset, \\ \arg \max_{j \in \mathcal{E}_t} F_j(Y_{t,j}), & \mathcal{E}_t \neq \emptyset, \end{cases}$$

where $F_j(y) = \Pr(Y_j \leq y)$ and $\mathcal{E}_t = \{j : \mathbb{I}\{Y_{t,j} > F_j^{-1}(q)\} = 1\}$ for some common high probability quantile level q . Each A_t can have state 0 corresponding to a *Quiet* state, where no exceedance is observed for any components of $\mathbf{Y}_t$, and where state j identifies component $Y_{t,j}$ is above the threshold and in *Active* state. We then feed the sequence as an input to the autoregressive transform and estimate the conditional probability $\hat{\pi}_\theta$ of the next token. Refer to [3] on transformer architectures.

For the second approach, let $\{\mathbf{Y}_t\}_{t=1}^N$ be a d -dimensional multivariate stochastic process, as in the first half of the session, now kept on the continuous scale: each margin is standardised to standard Laplace $\{\mathbf{X}_t\}_{t=1}^N$ and each observation written in polar form, $\mathbf{X} = R \mathbf{W}$ with $R = \|\mathbf{X}\|$ and $\|\mathbf{W}\| = 1$. Rather than scoring every day into a token, we model the extreme observations themselves as a *marked Hawkes process* in which every event is extreme,

$$\lambda(t) = \mu + \sum_{T_i < t} \eta \beta e^{-\beta(t-T_i)},$$

$$\mathbf{W} \mid \mathbf{W}_{\text{parent}} = \mathbf{w} \sim \text{vMF}(\mathbf{w}, \kappa), \quad R \mid \mathbf{W} = \mathbf{w} \sim \text{truncGamma}\{\alpha, g(\mathbf{w})\} \text{ above } u_q(\mathbf{w}),$$

where μ is the rate of the events that arrive on their own, $\eta < 1$ the mean number of children each event triggers and β the decay of their $\text{Exp}(\beta)$ delays, while g is the *gauge* of the limit set $G = \{\mathbf{x} : g(\mathbf{x}) \leq 1\}$ and $u_q(\mathbf{w}) = C_q/g(\mathbf{w})$ the q -quantile of the untruncated radial law in direction $\mathbf{w}$, so that an observation is extreme exactly when it falls outside $C_q \partial G$ [8]. A *cascade* is one such spontaneous event together with all of its descendants: a family rather than a time window, whose parenthood is never observed. We then feed the sequence of events to an autoregressive transformer carrying one head per factor of $p(T_{i+1}, \mathbf{W}_{i+1}, R_{i+1} \mid \mathcal{H}_i)$, read the estimates $(\tilde{\eta}, \tilde{\beta}, \tilde{\kappa}, g_\psi)$ off the trained network, and use them to recover the latent cascades and to generate new ones. Refer to [4] on Hawkes processes and [8] on the geometric approach.

Exercises

Exercise 1: Autoregressive Transformer model for cascading extremes

Open `01_gamex_cascading_extremes_autoregressive.ipynb`.

1. Simulate the five-dimensional heavy-tailed VAR(5) process with time-varying volatility. Examine the simulated series and identify periods of volatility clustering and cascading extremes.
2. Tokenize the continuous process using the threshold $q = 0.98$ and the fitted generalized Pareto tails. Inspect the resulting token frequencies.
3. Fit the transformer to the categorical sequence $p_\theta(A_{t+1} \mid A_{t-L+1:t})$, and examine the training and validation negative log-likelihood. Compare its performance against the IID-token baseline.
4. Assess whether the fitted Transformer has learned the dependence between extreme events by comparing the observed and fitted one-step transition probabilities $P(A_{t+1} = j \mid A_t = i)$.
5. Generate multiple future cascades from a fixed observed history and compare the transformer's conditional token probabilities with those obtained directly from the true.

Exercise 2: A continuous Transformer

Open `02_gamex_continuous_transformer.ipynb`. The data are thirty-eight years of S&P 500 daily returns and the model has 51,842 parameters.

1. Inspect the windows and the shifted input–target pairs, and relate them to the factorisation $f(x_1, \dots, x_{17}) = \prod_t f(x_t \mid x_{<t})$.
2. Identify the density head and the Gaussian negative log-likelihood, then train and read the validation loss against the fixed-Gaussian baseline.
3. Follow the fitted σ_t through 2006–2010, then compare fans of futures generated from a calm day and from the worst day of the COVID crash.
4. Complete the GamexCoin activity: give the same network a Student- t head, rerun the four-sigma check, and compare the largest move each head generates.

Exercise 3: A marked Hawkes process of extremes

Open `04_gamex_hawkes_cascades.ipynb`.

1. Identify μ , η , β , κ , α , q and the gauge g in the code, and watch the extremes appear outside the limit set as n grows.
2. Build the `gx.Process`, check that no simulated size ever falls below $u_q(W)$, and watch the process unfold in time.
3. Verify the branching theory, $1/(1 - \eta) = 4.55$, then compare the EM estimates and the gauge likelihood with the truth. Why is there no threshold parameter to estimate?
4. Complete the GAMEX exchange activity: simulate a contagion weekend with η close to 1, then hand the times alone to EM and see how many of your cascades it extracts.

Exercise 4: Estimating and generating real cascades

Open `03_gamex_cascade_estimation.ipynb`. Training takes about two minutes with `EPOCHS = 30`.

1. Follow the three indices onto Laplace margins and identify the extreme days as the points outside the fitted surface $\{\hat{u}_q(w) w\}$.
2. Locate the four heads in `gamex_transformer.py` and match them to the factorisation of the next event; then train and compare $\tilde{\eta}$, $\tilde{\beta}$ and $\tilde{\kappa}$ with the classical seed.
3. Recover the latent cascades at those estimates and check that the largest ones are crises you recognise.
4. Generate a crisis event by event from a starting day of your choice, then read the three diagnostics: cascade size against $1/(1 - \tilde{\eta})$, the severity staircases and the exceedance PIT.

Questions for Discussion

- How does different scoring rules affect the sequence of extremal tokens?
- Which diagnostics would make you trust generated cascades for risk estimation: the prediction bands, the PIT, the mark laws, the limit set, or some combination?
- What is gained by modelling the extreme days directly as a self-exciting process, rather than fitting a model to all days and keeping only the extremes of its simulations?
- How would the analysis change with ten indices instead of three? Which parts of the pipeline carry over directly, and which do not?

References

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